

4 (**short**) Figure 4 shows a semi-circular arch of radius R , with frictionless pin-joints at A, B and C. B is at the midpoint of the arch AC. A force F is applied to the arch as shown, at 45° to the vertical, mid-way between B and C. Explain why the reaction at A acts along the line AB and draw the bending moment diagram for this loading on a sketch of the arch, marking salient points and using the convention that the bending moment is plotted on the “tensile side” of a member. [10]

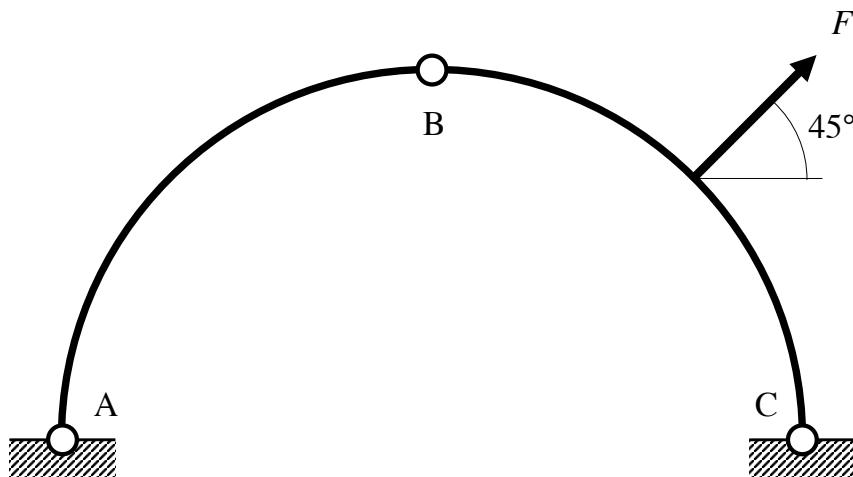


Fig. 4